



LombardTokenizer: Disentanglement and Control of Vocal Effort in a Neural Speech Codec

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Abstract

Disentangling distinct types of information in speech representations is crucial for improving speech synthesis and voice conversion systems. In this work, we introduce LombardTokenizer, a neural speech codec able to separate features related to vocal effort from other acoustic (and semantic) information. This model is built on SpeechTokenizer, a model proposed in the literature based on multi-stage quantisation, which focused on isolating semantic content in its first quantisation layer. We show that the level of vocal effort can be effectively captured in the second quantisation layer by conditioning the quantisation layer with neural encoders trained to represent vocal effort. Experimental results demonstrate that the proposed method significantly outperforms existing methods in speech conversion between neutral and Lombard speech, while maintaining excellent speech synthesis quality, offering improved control over vocal effort and naturalness of synthesised speech.

Index Terms: neural speech codec, lombard speech, vocal effort encoding, latent space regularisation, speech tokenization

1. Introduction

Neural audio codecs have revolutionised speech processing by enabling efficient compression and high quality audio synthesis. At the core of state-of-the-art neural codecs, such as SoundStream [1] and EnCodec [2] is Residual Vector Quantisation (RVQ) [3], a cascaded quantisation architecture that hierarchically represents complex speech and audio information, from coarse global features to fine-grained acoustic details. This hierarchical structure has proven effective for both speech manipulation [4] and low-bitrate coding [5], paving the way for more advanced applications in speech synthesis and transformation.

Building on the success of RVQ-based neural codecs, SpeechTokenizer [6] introduced a novel approach to disentangling information in speech signals. The model's first RVQ layer is explicitly guided by the output of a self-supervised learning (SSL) speech model, namely HuBERT [7], ensuring that it captures what the authors define as semantic representations. This effectively isolated phonetic information from other source of acoustic variability encoded in the subsequent RVQ layers, allowing for separate modifications to phonetic content and acoustic representations in voice conversion applications. However, disentangling different dimensions of acoustic variations remains an open challenge, particularly when analysing speech produced at varying levels of intensity and vocal effort. Such variations in vocal effort can be observed in challenging situations of communication, such as speaking over a distance, in noisy environments (a phenomenon known as the 'Lombard effect' [8]) or when addressing to a hearing-impaired individual. In all cases, increased vocal effort is typically accompanied

by global acoustic modifications such as an elevated fundamental frequency (f_0), a flattened spectral slope and an increased frequency of the vowel's first formant (F1) [9, 10], but also by segment-specific modifications [11, 12, 13]. Additionally, a form of hyper-articulation can be observed in some of these speech styles, such as Lombard speech (i.e., produced in noise), with enhanced phonemic contrasts, prosodic cues and voice modulations [14, 15] contributing to their increased speech intelligibility under adverse conditions [16].

Traditional approaches to neutral-to-Lombard speech conversion often rely on explicit manipulation of acoustic features, such as pitch contours or energy levels [17, 18, 19, 20]. More recent methods use deep neural networks to learn a mapping between neutral and Lombard speech, either in a supervised or unsupervised manner [21, 22, 23, 24]. In parallel, research on voice conversion has also explored techniques for disentangling speaker identity and speaking style from linguistic content. Models like FreeVC [25] combine the SSL model WavLM [26] with d -vector speaker embeddings [27] to achieve high quality voice conversion. Other works, such as [28] integrate HuBERT with a pitch encoder and a speaker encoder to model semantic content, prosodic information, and speaker characteristics separately. These approaches demonstrate the effectiveness of SSL models in generating semantic representations and disentangling them from prosodic and speaker features in neutral speech. However, they do not explicitly target the disentanglement of specific prosodic and acoustic features associated with the Lombard effect.

This paper introduces LombardTokenizer, a neural speech codec designed to encode speech modifications induced by variations in vocal effort separately from other acoustic and semantic information. This model is built on SpeechTokenizer and its RVQ structure. In addition to isolating the semantic content in the first RVQ layer, we demonstrate that Lombard speech transformations can be effectively captured in the second RVQ layer by leveraging deep neural encoders trained to represent vocal effort. Notably, we train the model on neutral speech, fine-tune it on loud speech and evaluate it on loud and Lombard speech following a zero-shot approach to assess its ability to generalise to unseen vocal effort conditions. A second contribution of this paper is to replace HuBERT with the multilingual SSL model mHuBERT [29] in the first RVQ layer to improve the semantic representation and enable zero-shot generalisation across languages. The performance of LombardTokenizer was assessed both objectively and subjectively from experiments conducted on a loud speech dataset (AVID [30]) as well as a new Lombard speech dataset (FLombard [31]) specifically recorded for this study, which can therefore be considered as the third contribution of this paper.

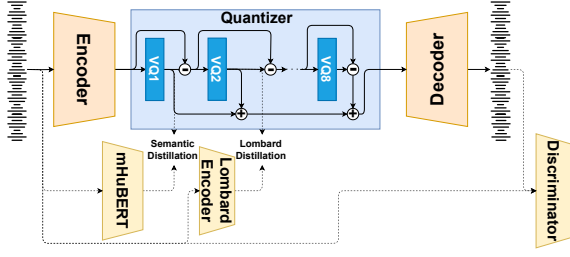


Figure 1: Schema of the proposed LombardTokenizer model.

2. Method

In this section, we first present the architecture of LombardTokenizer. We then describe the two datasets with variation of vocal effort. Finally, we present the training procedure designed to disentangle vocal effort and semantic information in our model.

2.1. Model Architecture

The proposed LombardTokenizer model is based on the architecture of SpeechTokenizer, as depicted in Fig. 1. It uses the convolutional-based (CNN) encoder-decoder network, with a two-layer BiLSTM at the end of the encoder and front-end of the decoder convolution blocks. The output of the encoder is quantised through eight RVQ layers. In SpeechTokenizer, the first RVQ layer is forced to learn semantic-oriented codebooks during training via distillation by HuBERT.

Our model introduces two key extensions that are detailed in Sec. 2.3: (i) Lombard-specific encoders plugged on the second RVQ layer to force the latter to learn codebooks encoding vocal effort features, and (ii) the replacement of HuBERT with the multilingual mHuBERT model [29] to generalise the disentanglement of semantic information in the first RVQ layer to several languages.

2.2. Speech Datasets

To extract vocal effort relevant information, we use the AVID dataset [30]. This dataset includes recordings of 50 non-necessarily native English speakers (25 males and 25 females) exhibiting controlled changes in four different intensity categories (soft, normal, loud, very loud). To elicit each category, the speakers were instructed with textual guidelines but were not forced to produce speech in a predefined intensity level. In this work, we used the speech signals produce during the sentence reading task, totalling 8 h of speech, with 2 h for each intensity condition and approximately 10 min per speaker. While the AVID dataset exhibits a high dynamic intensity range (40 dB), it does not explicitly capture natural Lombard speech as it occurs in real noisy environments.

To address this limitation, we introduce FLombard [31], a dataset where speakers actively adapt their speech in response to background noise, ensuring a more authentic representation of Lombard speech characteristics. This newly collected dataset features 40 French speakers (20 male and 20 female) producing Lombard speech under four different levels of white noise heard via headphones (0, 65, 75, 85 dB SPL). The recordings were conducted in a controlled listening room to ensure high quality audio capture. In total, the dataset contains approximately 8 h of speech, with 2 h for each noise condition and an average of 13 min per speaker.

2.3. Training Strategy

Lombard Speech Encoding: To capture vocal effort features, we first introduce two different Lombard speech encoders, E_1 and E_2 , and train them on the AVID dataset. As for E_1 , we implemented a d -vector model from [32], pre-trained on LibriSpeech and VoxCeleb [33]. This model takes as input a 40-band mel-spectrogram and outputs a speaker embedding vector of size 256. The model is optimised with the generalised end-to-end loss function [27]. To make the model more suitable for encoding vocal effort variation, we concatenate the mean intensity of the speech signal with the mel-spectrogram input and apply a 41-to-40 fully-connected (FC) layer during the fine-tuning on AVID. The second model, E_2 , is our own. It consists of three CNN layers, followed by an LSTM layer and two FC layers. It was trained on AVID to predict the intensity of a normalised input waveform (whose RMS is 1). The intensity to predict was made speaker-independent: we applied a Min-Max normalisation for each speaker in the AVID dataset to scale the mean intensity of all signals between 0 and 1. After training, instead of using the final output of E_2 , we extract the intermediate representation after the first FC, with a dimensionality of 32, to capture vocal effort features.

Semantic and Lombard Distillation: To disentangle semantic and vocal effort from other acoustic features in LombardTokenizer, we take advantage of the hierarchical structure of the RVQ module. We use the SSL model mHuBERT and one of our two Lombard speech encoders to guide the first and the second RVQ layers, respectively, during LombardTokenizer training, as shown in Fig. 1. The mHuBERT, E_1 , and E_2 representations have a dimensionality of 768, 256 and 32, respectively. To match those dimensions with the quantised output of the first and second RVQ layer (each of size 1024), we apply a 1024-to- D FC layer to the quantised features before applying the distillation loss, following the approach described in [6]:

$$\mathcal{L}_{distill} = -\frac{1}{D} \sum_{d=1}^D \log \sigma \left(\cos \left(\mathbf{A} \mathbf{Q}^{(:,d)}, \mathbf{X}^{(:,d)} \right) \right), \quad (1)$$

where $\mathbf{Q}$ denotes the quantised output of the first or second RVQ layer, $\mathbf{X}$ denotes the corresponding semantic or Lombard speech representation, $\mathbf{A}$ is the projection matrix, and D is the dimension of the representation (768, 256, or 32). The superscript $(:, d)$ denotes a vector comprising values from all time steps at dimension d , while $\cos(\cdot)$ is the cosine similarity and $\sigma(\cdot)$ denotes the sigmoid activation. Those two losses comes in addition to the reconstruction, discriminative and RVQ commitment losses, necessary to train SpeechTokenizer [6].

Since we have two possible Lombard speech encoders to plug, E_1 and E_2 , this gives us two versions of LombardTokenizer, that we call LT₁ and LT₂, respectively. In both versions, the representation distillation strategy ensures that the second RVQ layer learns to encode features related to vocal effort, while the first layer remains focused on semantic information.

Fine-tuning on Vocal Effort: We first pre-trained our LombardTokenizer models with semantic and Lombard distillations on a large-scale, multi-speaker dataset, using LibriSpeech dataset [34]. Then, we further fine-tuned the model on the AVID dataset. Fine-tuning on data that encompass vocal effort variation is a crucial step in our approach, as it allows the model to explicitly learn and encode the acoustic characteristics associated with different levels of vocal effort. Unlike pre-trained speech tokenizers that primarily capture general speech features, fine-tuning on such datasets enables the codebook of the

second RVQ layer to effectively encode variations in intensity, spectral tilt, and prosodic cues linked to vocal effort.

2.4. Experiments

Details of the models implementations and examples of synthesised speech are available in our demo website.¹

Synthesis quality: A first step was to compare the quality and intelligibility of speech synthesised by LombardTokenizer against two strong baselines: EnCodec [2], a state-of-the-art neural audio codec, and SpeechTokenizer [6], the backbone of our method. Those four models were trained on LibriSpeech 360h ‘clean’ training subset and fine-tuned on 6h from 40 speakers of AVID. The sampling rate of the speech signals was 16 kHz. For training and fine-tuning, we used 2 A100 GPUs, with a learning rate of $1e^{-4}$, for 20 epochs and a batch size of 32 per GPU. Evaluation of speech synthesis was performed on unseen parts of LibriSpeech and AVID, with 2h for 2620 sentences and 1.50h for 2200 sentences, respectively, as well as on FLombard, with 1.30h for 1920 sentences. Two metrics, PESQ [35] and STOI [36], were used to evaluate objectively the quality and intelligibility of the speech synthesized.

Modeling and conversion of vocal effort: In a second step, we evaluated the ability of LombardTokenizer to capture speech transformations related to vocal effort and control this dimension in re-synthesis. An evaluation combining both objective metrics and perceptual listening tests was conducted on both AVID and FLombard datasets to evaluate performance under in-domain and zero-shot conditions, respectively. The evaluation compares the two variants of LombardTokenizer (LT₁ and LT₂) against FreeVC [25], a state-of-the-art voice conversion model that also combines an SSL model (WavLM [26]) and a speaker embedding. Two variations of FreeVC were considered: one using the standard speaker encoder from [32] and one modified with the Lombard speech encoder E_1 , referred to as FVC_{spk} and FVC_{lmb}, respectively. Both models were trained and fine-tuned with a procedure similar to LombardTokenizer. Vocal effort conversion was performed by switching the vocal effort encoding of a source and a target signals. For LombardTokenizer, the output of the second RVQ layer of an encoded source signal was replaced by the output of the second RVQ layer of an encoded target signal, while keeping the other layers unchanged. For FreeVC, the embedding (speaker or Lombard) of the encoded source signal was replaced by the embedding of an encoded target signal, while keeping the output from WavLM unchanged. Vocal effort adaptation across speakers is flexibly achieved by both methods, by selecting a target representation from either the same speaker (intra-speaker conversion) or a different speaker (inter-speaker conversion).

An objective evaluation of these four conversion methods was conducted on the following data: for AVID and FLombard dataset, 24 utterances were selected, each produced by 8 speakers (4 males and 4 females) at four levels of vocal effort. Each sentence was then converted to the three other levels of vocal effort with our four models, given either a target signal with same speaker and same utterance (intra-speaker conversion), or two target signals with two different speakers and same utterance (inter-speaker conversion). This resulted in 27 648 sentences to evaluate for each dataset. Two metrics, the Word Error Rate (WER) and Equal Error Rate (EER), were used to quantify content preservation and speaker similarity, respectively. Additionally, the Jensen-Shannon divergence (JS), a symmetrised

Table 1: PESQ and STOI scores for EnCodec (EC), SpeechTokenizer (ST) and LombardTokenizer (LT) reported for LibriSpeech (LB), AVID and FLombard (FL) test sets. We report mean scores and 95% confidence intervals.

Dataset	Method	PESQ [−0.5; 4.5] ↑	STOI [0; 1] ↑
LB	EC	3.32 ± 0.03	0.94 ± 0.01
	ST	3.01 ± 0.05	0.93 ± 0.02
	LT	3.07 ± 0.04	0.93 ± 0.02
AVID	EC	3.25 ± 0.07	0.92 ± 0.03
	ST	2.88 ± 0.08	0.91 ± 0.04
	LT	3.01 ± 0.06	0.91 ± 0.05
FL	EC	2.94 ± 0.11	0.90 ± 0.06
	ST	2.54 ± 0.11	0.88 ± 0.07
	LT	2.81 ± 0.07	0.89 ± 0.05

version of the Kullback-Leibler divergence, was computed on frame-wise fundamental frequency and spectral slope distributions (JS-F0 and JS-Slope) to evaluate acoustic consistency between the transformed and target signals.

Vocal effort modification was also assessed through a listening test with comparative ratings. The latter consisted in rating the perceived vocal effort on a scale from 0 (soft) to 100 (very loud) of 8 signals: the four natural rendition of an utterance with the four levels of vocal effort, and the conversion of vocal effort of the utterance from a source to a target level with the four models, at non-adjacent levels of vocal effort (i.e., between soft and loud levels, or between normal and very loud levels). A single {source, target} pair was selected for each utterance, and combinations were randomly distributed across utterances. For each dataset, the same 24 utterances were rated with four speakers and both the intra- and one inter-speaker conversion, for a total of 192 comparative ratings. 40 English speakers for AVID and 40 French speakers for FLombard, all native, were recruited via the Prolific platform² and performed 24 comparative ratings each, following a Latin Square design. Statistical significance between score distributions across models and vocal effort levels was assessed using Dunn’s test.

3. Results

3.1. Synthesis quality

Table 1 presents the comparison of synthesis quality and intelligibility – as evaluated through the PESQ and STOI scores – between our method: LombardTokenizer (LT), and two baselines: EnCodec (EC) and SpeechTokenizer (ST). As both LombardTokenizer variants present no significant difference in speech synthesis performance, we only report the results for LT₁. The results show that for neutral speech – as represented in the LibriSpeech dataset (LB) –, LombardTokenizer and SpeechTokenizer achieve similar levels of synthesis quality and intelligibility, which reflects their shared architecture and strategy of semantic regularisation. For speech produced with increased vocal effort – as represented in AVID and FLombard (FL) datasets –, LombardTokenizer outperforms SpeechTokenizer, with the greatest performance gap observed for the FLombard dataset, corresponding to a zero-shot generalisation scenario. This supports the interest of using multilingual SSL and Lombard speech encoder to guide the RVQ layers. EnCodec consistently achieves the highest scores of synthesis quality and intelligibility, certainly because it does not constrain its quantised layers to capture specific types of information. Interestingly,

¹<https://github.com/lombardtokenizer/lombardtokenizer.github.io>

²<https://www.prolific.com>

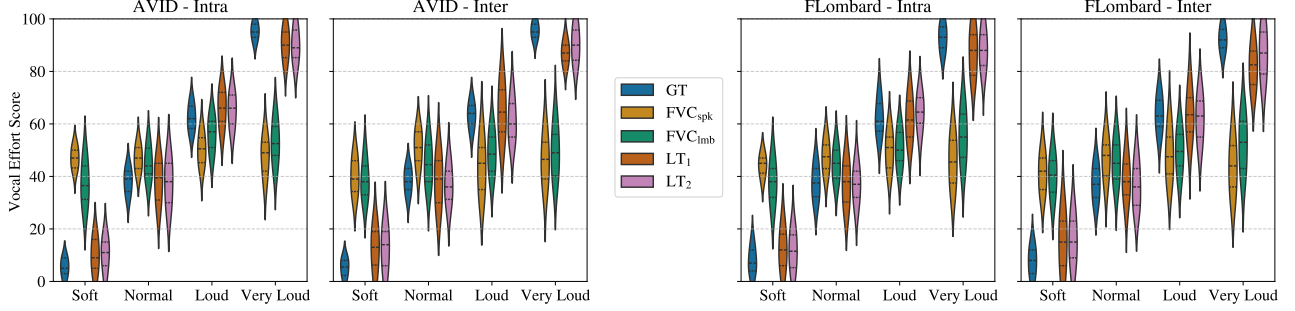


Figure 2: Perceptual evaluations of vocal effort conversion with the two variants of LombardTokenizer and FreeVC on the AVID and FLombard test sets (intra- and inter-speaker conditions).

this advantage becomes less pronounced with the FLombard dataset, which includes expressive speech and unseen data conditions. In these cases, LombardTokenizer demonstrates competitive performance due to its targeted regularisation strategies.

3.2. Modeling and conversion of vocal effort

Objective evaluation:

As shown in Table 2, both variants of LombardTokenizer outperform the FreeVC models across all objective metrics on both datasets. For the AVID dataset, LT₁ achieves the lowest WER (10.97%) and EER (6.67%), indicating strong performance in maintaining both intelligibility and speaker identity. Meanwhile, LT₂ slightly outperforms LT₁ in terms of JS-F0 (0.03) and JS-Slope (0.07), reflecting its superior ability to capture the spectral modifications that accompany variations in vocal effort. The performance gap becomes more pronounced for the FLombard dataset. Here, both LombardTokenizer variants maintain their advantage, with LT₂ achieving the lowest WER (15.77%), which demonstrates its robustness in unfamiliar conditions. While EER scores increase slightly compared to AVID, reflecting the added challenge of vocal effort conversion on unseen conditions, LT₁ and LT₂ still outperform both FreeVC models. This suggests that they retain speaker characteristics more effectively, even under zero-shot conditions.

Perceptual evaluation:

The results from the perceptual evaluation, displayed in Fig. 2, confirm that the listeners were able to perceive and distinguish different levels of vocal effort. Indeed, for the untransformed samples of both datasets (Ground Truth - GT), the perceived level of vocal effort shows significantly distinct distributions for each category of produced vocal intensity ($p < 0.01$). This distinctiveness is slightly less pronounced in FLombard compared to AVID, which may stem from the greater vocal intensity range covered in AVID (spans a 40dB range between soft and very loud speech) compared to FLombard (only 20 dB).

Conversely, regardless of the dataset or speaker configurations (intra- and inter-speaker), the perceived effort of samples transformed with FreeVC does not exhibit significant differences across the four categories of produced vocal intensity. This suggests that FreeVC does not manage to generate distinct variations in vocal effort. While such result was somewhat expected for the variant using the standard speaker encoder, it is more surprising for the FreeVC model that relies on the Lombard encoder, indicating its poor ability to leverage Lombard-specific information for expressive conversion. In contrast, both LombardTokenizer variants appear to accurately convert vocal effort, showing similar distributions of perceived effort between the transformed samples (after conversion) and

Table 2: Objective evaluations of vocal effort conversion with the two variants LombardTokenizer and FreeVC on AVID and FLombard. We report WER, EER, JS-F0 and JS-Slope.

Dataset	Method	WER (%) ↓	EER (%) ↓	JS-F0 ↓	JS-Slope ↓
AVID	GT	7.83	4.79	—	—
	FVC _{spk}	20.35	16.67	0.24	0.34
	FVC _{lmb}	17.50	25.00	0.22	0.33
	LT ₁	10.97	6.67	0.04	0.08
	LT ₂	12.73	7.83	0.03	0.07
FL	GT	11.46	7.34	—	—
	FVC _{spk}	24.04	22.50	0.16	0.30
	FVC _{lmb}	32.63	33.13	0.14	0.28
	LT ₁	17.74	16.88	0.05	0.06
	LT ₂	15.77	18.29	0.04	0.04

the untransformed ones (corresponding to the conversion target, GT), across all four categories of vocal intensity. A slight performance drop is observed in inter-speaker conversion (evidenced by wider distributions), compared to the intra-speaker conversion. However, this difference remains non-significant, indicating that LombardTokenizer maintains robust conversion quality across different speakers.

The results of both objective and perceptual evaluations confirm the superiority of our approach. Indeed, while both LT₁ and FVC_{lmb} utilise the same Lombard encoder, only our model achieves effective vocal effort conversion. Moreover, the similar performances observed between the two variants, LT₁ and LT₂, demonstrate the strong ability of the LombardTokenizer method to generalise to various vocal effort representations.

4. Conclusion

This paper introduced LombardTokenizer, a novel framework that extends the concept of RVQ regularisation by introducing targeted regularisation in the second RVQ layer using Lombard speech encoders, while leveraging the multilingual mHuBERT for semantic guidance in the first RVQ layer. Two variants of LombardTokenizer were explored: one based on a d -vector encoder and another using an intensity predictor. Both objective and perceptual evaluations demonstrated that LombardTokenizer outperforms existing models in both speech synthesis and conversion tasks. Notably, LombardTokenizer achieved strong performance regardless of the vocal effort representation and demonstrated effective zero-shot generalisation. More generally, this work provides new insight to the problem of disentangling different dimensions of speech variation (such as vocal effort, expressivity, speech clarity) in neural speech codecs.

5. References

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